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30 April 1990
SUPERSEDING
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MILITARY SPECIFICATION

SERVO DRIVE, TRAVERSE: 10548052

This specification is approved for use within the U.S. Army Armament,Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification **covers the** traverse servo drive which is a part of the Antenna Assembly AS-2049/VPS-2 or used with Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications. standards and handbooks form a part of this **specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.**

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control, Material, General Specification Governing the Manufacture and Inspection of
MIL-S-20708	Synchros, General Specification for
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, PicatinnY Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD *Form 1426)* appearing at the end of this document or by letter.

AMSC N/A
FSC 1285
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MIL-S-45743 - Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment
MIL-S-46844 - Soldering, Bath Soldering of Printed Wiring Assemblies
MIL-A-50687 - Antenna Assembly AS-2049/VPS-2: 10556312

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified,the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548053 Housing Traverse
10548052 Traverse Servo Drive
10548193 Servo Electronics
10548326 Traverse Servo Drive Runs List, Wire
10548389 Antenna Schematic
10548395 Traverse Trunnion
10556312 Antenna Assembly, AS-2049/VPS-2

PACKAGING DATA SHEET

P10548052 Packaging of Traverse Servo Drive

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except **for** associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and **assemblies** shall comply with all requirements specified on Drawing 10548052, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the traverse servo drive shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. **Part identification and marking.**

d. Workmanship.

e. Soldering electrical connections.

3.3 Environmental service conditions.

3.3.1 Vibration. The traverse servo drive shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 **hertz**	**0.06 inches double amplitude**
15 to 25 hertz	**0.04 inches double amplitude**
25 to 55 hertz	**0.02 *inches*** double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ± 10%.

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3.3.2 Shock. The traverse servo drive shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks, consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The traverse servo drive shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Non-operating. The traverse servo drive shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The traverse servo drive shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between –40 deg. F and +155 deg. F.

3.3.4.2 Non-operating (storage). The traverse servo drive shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg. F for a period of 24 hours and +160 deg. F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating] The traverse servo drive shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for period of ***10*** days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the traverse servo drive.

3.5 Performance.

3.5.1 Component resistance. After 10 minutes of operation, the resistance of components with traveling elements to fixed elements shall be as follows:

a. Drive, motor resistance: 0.75 to 10 ohms.

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b. Tachometer generator: 350 ± 100 ohms.
Synchro stator S1 to S2 or S3: 500 ± 100 ohms.
c. Synchro rotor R1 to R2: 320 + 80 Ohms.
e. Drive brake: 200 ± 5O ohms with diode cathode connected to terminal J1-n.

3.5.2 Servo drive brake. When ***+26 ± 2 volts*** (V) is applied between J1-n and J1-a, the servo drive brake shall release.

3.5.3 Drive motor. When conditions of 3.5.2 are present, the drive motor shall begin to operate between 0.3 and 3.0 volts direct current (Vdc) of either polarity.

3.5.4 Tachometer. When the drive voltage of 3.5.3 is increased to 10 v the tachometer (generator) shall produce 13.0 ± 4.0 V and shall be steady within 1.0 v.

3.5.5 Synchro. The synchro (Bl) shall be aligned to electrical zero to within 1/8 deg.

2

3.5.6 Potentiometer. The /\ potentiometer shall provide a null of less than 120 millivolts (mv) at O deg. traverse lead angle and shall provide 2.4 ± 0.4 V at ± lO-deg. , when excited with 15 Vdc .

3.5.7 Limit switch. A limit switch shall operate at ± 26.5 ± 0.5 deg. traverse lead angle.

3.5.8 Gearing. There shall be no evidence of backlash in the data train gearing.

3.5.9 Servo drive dynamic test. The traverse servo drive after having met the requirements of 3.5 through 3.5.8 shall meet the dynamic test **as** specified in MIL-A-50687 when substituted into an Antenna AS-2049/VPS-2. Antenna AS-2049/VPS-2 and test facilities shall be as specified in MIL-A-50687. When the traverse servo drive is procured as a spare item, the antenna (10556312) and servo electronics (10548193), referenced in MIL-A-50687, will be supplied to the contractor **as** GFP.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical p~~r-~~ (see 4.3).

3.7 Workmanship. The workmanship requirements of MIL-F-13926, MIL-S-45743 and MIL-S-46844 shall apply.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5 and all applicable drawings. The **inspection set forth in this specification-shall** become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c* Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 °F to +90 °F'
b.	Relative Humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg

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4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) traverse servo drive for inspection in accordance with 4.3.2.

4.3.2 **Inspections to be performed. First article assemblies, components and** test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I and Table II herein and to any or all requirements of the applicable drawing.

4.3.3 Rejection. If any assembly, component ***or*** test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate Its inspection upon failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior ***or*** subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revision . All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination and tests.

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4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the technical data package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item **or** lot.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL :	None	
MAJOR :		
101 Electrical	3.4	4.7.2
102 Performance	3.5	4.7.3
MINOR :	None	

4.5 Special sampling. One traverse servo drive out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

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TABLE II. Special sampling, environmental.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	None	
MAJOR:		
103 Vibration	3.3.1	407.1.1
104 Shock	3.3.2	4.7.1.2
105 Altitude	3.3*3	4.7.1.3
106 Temperature	3*3.4	4.7.1.4
107 Humidity (nonoperating)	3.3*5	4.7.1.5
MINOR:	None	

4.5.1 Failure of sample. Should **any** one item of a special sampling fail to meet-any of the-specified test requirements, acceptance of the product shall be-suspended by the Government **until** necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIT-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test

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equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the standard test equipment listed in Table III shall be utilized in performing the applicable test methods and procedures specified in 4.7.

TABLE III. Test equipment list.

Item Number		Description
	Servo drive tester	See Figure 1. See Table IV for parts list.
2.	Dc power supply	24 Vdc, 0.5 A. Load regulation ± 1%. Ripple: Less than 100 mV.
3.	Dc power supply	0 to 30 Vdc, 10 A. Load regulation: ± 1%. Ripple: Less than 100 mV.
4*	Dc power supply	15 Vdc, 100 mA. Load regulation: ± 0.1%. Ripple: Less than 10 mV.
5.	Aligning fixture for elevation assembly	See Figure 2.
6.	Oscilloscope	Sensitivity: 0.1 V/cm. Frequency response: 100 kHz. Time base: 1 to 100 ms/cm.
7.	Ac-dc voltmeter	Range: 500 mV to 150 V. Input impedance: 10 megohms.

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TABLE III. Test equipment list (continued) .

8.	Inspection mirror	Lighted, flexible inspection mirror.
9.	Ohmmeter	R x 1 scale: O to 1 k *Ohm.* R x 100 scale: o to **100** k *Ohm.* Short test current R *x 1* scale: Not more than 200 mA. Short test current R x 100 scale: Not **more than 20 mA.**

TABLE IV. Servo drive tester parts list.

Designation	Nomenclature e	Part number
S1	SPST toggle switch	MS35058-22
TP1-TP24	Banana jack	H. H. Smith DF30
B1	Control transmitter synchro dial assembly or equivalent.	23CX4 per MIL-S-20708 with Theta Instruments, Model DA-23-2405
R1	Resistor	RCO7GF1O3J
cl	Capacitor	TRW capacitor X663F, 1 F, 100 WVdc or equivalent.
J1	Connector	Bendix PT06A-20-39S or equivalent.
DS 1	Pilot lamp	MS25257-4

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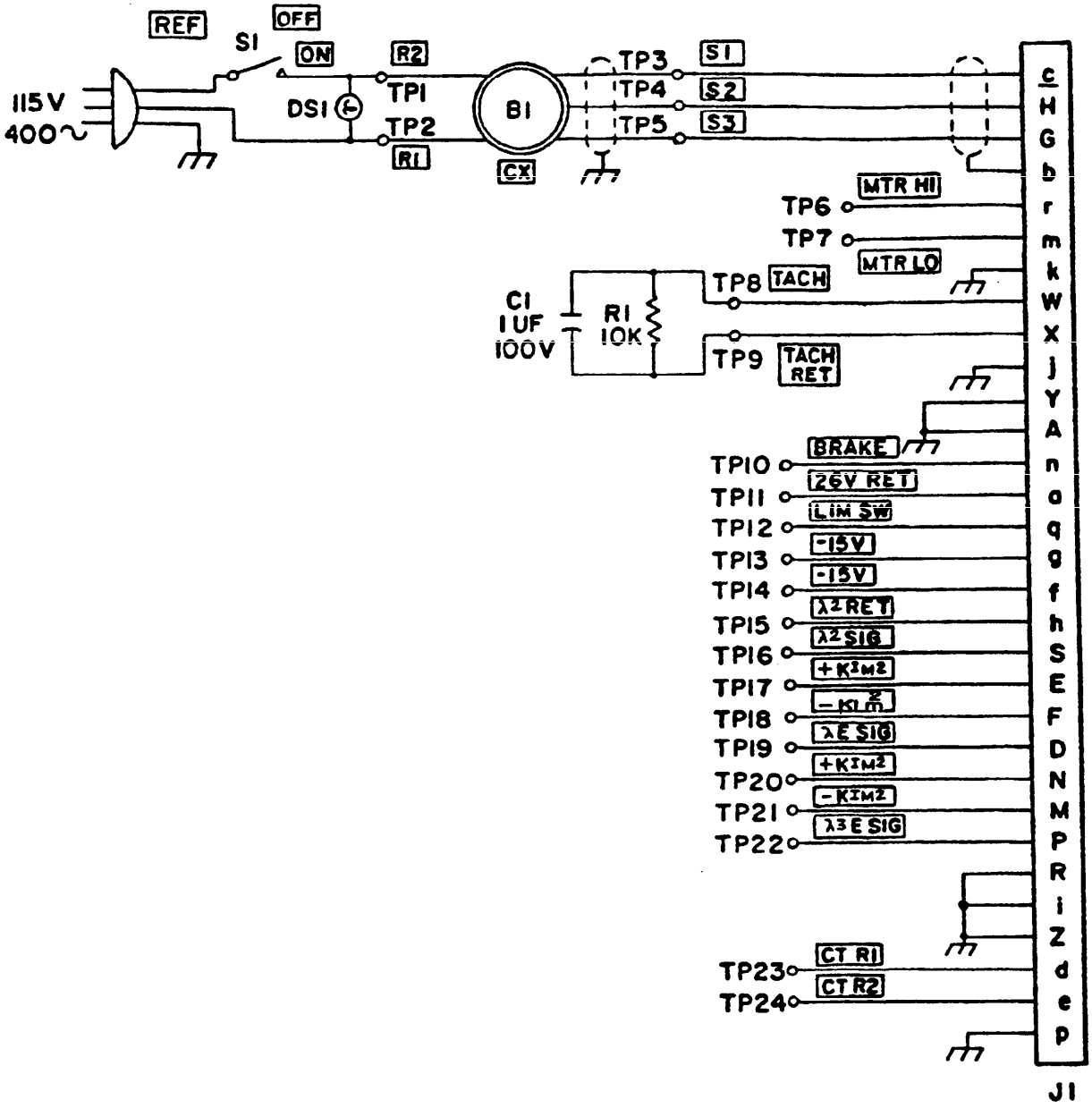


FIGURE 1. Servo drive tester.

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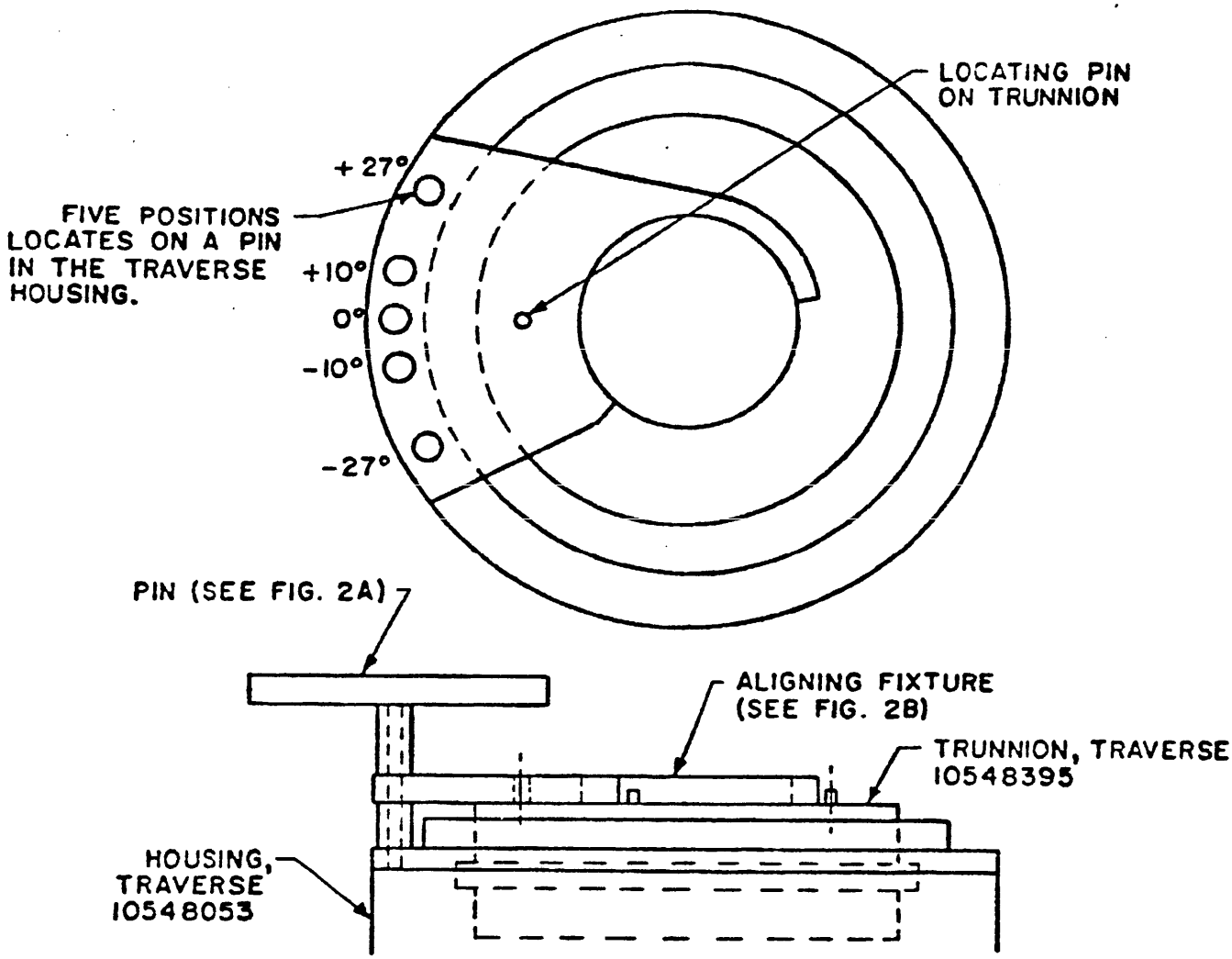


FIGURE 2. Aligning fixture for traverse assembly.

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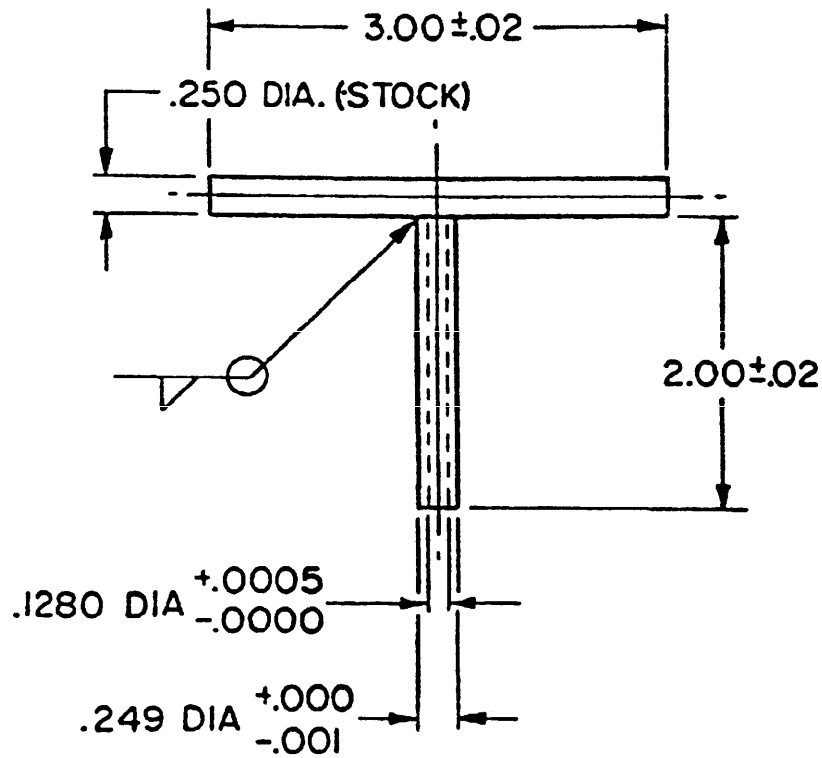


FIGURE 2A. Pin aligning fixture.

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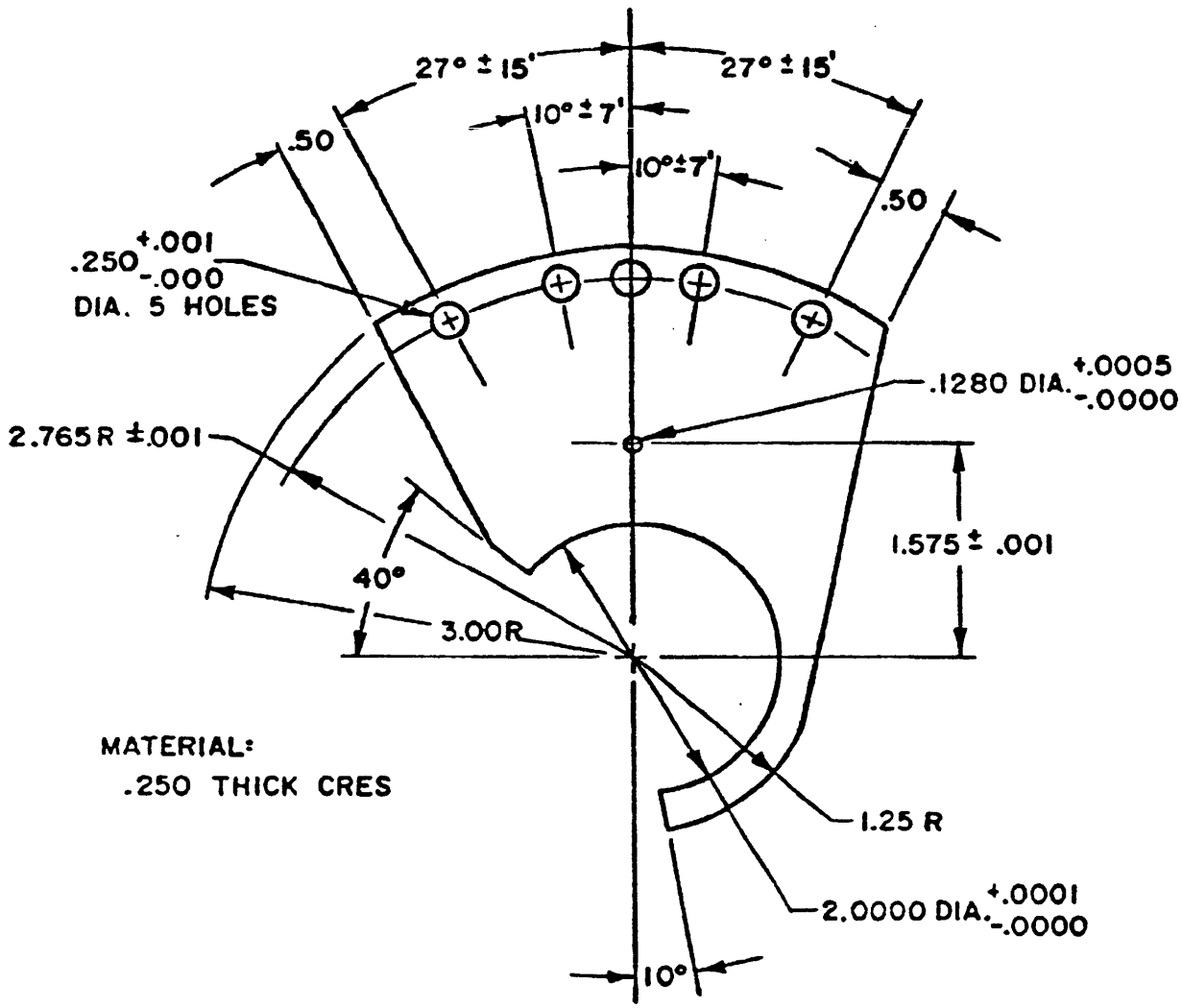


FIGURE 2B. Traverse aligning fixture.

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514.3, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I, only steps 4 through 8 of paragraph II-2.1 apply. The traverse servo drive shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the traverse servo drive shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock . This test shall be performed as specified in Procedure I,Section 11 of Method 516.3 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at **5.5** milliseconds + 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the traverse servo drive shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500.2, of MIL-STD-810. The operation checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

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4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating] This test shall be performed as specified in procedures I and 11, Method 501.2, of MIL-STD-81O with constant temperature extremes as specified in **3.3.4.** The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low **(operating and non-operating). This test shall be performed as specified in procedures I and II, Method 502.2, of** MIL-STD-81O with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed as specified in procedure III, Method 507.2, of MIL-STD-81o except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for for compliance with 3.3.5.

4.7.2 Electrical. Check the wiring visually using schematic Drawing 10548389 and runs list Drawing 10548326 as a reference. Continuity shall be inspected by standard measuring equipment, item 9. for compliance with 3.4.

CAUTION: When measuring potentiometers with an ohmmeter, set ohmmeter to X 100 scale to start, and then switch to Xl scale after a short is measured on X 100 scale. This will prevent damage to the function potentiometers.

4.7.3 Performance. The selected traverse servo drives shall be tested as follows:

4.7.3.1 Brake activation. Check that REF switch on servo drive tester is in the OFF position. Connect the test cable between the tester **and the** unit under test. Connect power supply, item 2, with positive lead to BRAKE (TP1O) and negative lead to 26 V (TP1l). Adjust for 24 V. Listen for brake actuation.

4.7.3.2 **Motor run-in. Connect servo drive tester to elevation servo drive. Connect a dc power supply, item 3, with positive terminal connected to MTR HI (TP6) and negative terminal connected to MTR LO (TP7). Turn on power supply and increase voltage until drive starts turning. Increase voltage to 10 V and allow** motor to run in alternating rotational directions for 10 minutes; motor direction to be reversed just prior to engagement of stop pin MS171434. Then, turn off the power supply and disconnect the test leads.

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4 .7.3.3 **Resistance** check.

a. J1-4 to J1-m: 0.75 to 10 ohms.

b. J1-r to J1-Y: Open.

c. J1-W to J1-X: 250 to 450 Ohms.

d. J1- c to J1-H, J1- c to J1-G and J1-G to J1-H: **400** Ohms to 600 ohms.

e. J1-d to J1-e: 240 to 400 ohms.

f. J1-Y to chassis: Short.

g" J1-A to chassis: Short.

h. J1-n to J1-a (with positive voltage on n): 150 to 250 ohms.

i. J1-n to J1-a (with positive voltage on a): Less than 100 ohms on Xl ohmmeter scale.

4.7 .3.4 Motor-generator.

a. Connect the dc power supply, item 3, with positive terminal connected to MTR HI (TP6) and negative terminal connected to MTR LO (TP7). Turn on output voltage supply. Increase voltage until drive starts turning. Applied voltage should be between 0.3 and 3 v. The drive should rotate clockwise with respect to the housing. Connect a VOM set to read dc voltage with negative lead on TACH (TP8) and positive lead on TACH RET (TP9).

b. Increase motor voltage to 10 V. Tachometer voltage shall be 9 to 17 V, of the polarity indicated in a.

c. Tachometer voltage shall be steady within 1 **V as the** unit is run.

d. **Turn off power supply. Reduce output voltage control to zero. Reverse** power **supply polarity. on supply and increase-voltage until drive starts turning. Applied voltage should be between 0.3 and 3 v.**

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e. The drive should rotate counterclockwise. Increase the motor voltage to 10 V. Reverse tachometer leads to obtain +9 to +17 v at TACH (TP8) with respect to TACH RET (TP9). Tachometer voltage shall be steady within 1 V as the unit is run. Turn off the motor power supply and disconnect the VOM.

4.7.3.5 Synchro.

a. **Install traverse assembly alignment fixture (see Figure 2) and move drive to 0 deg., using motor power supply . Adjust CX dial for exactly 0 deg.**

NOTE : **CX** of servo drive tester to be zeroed in accordance with MIL-S-20708.

b. Connect a jumper from CX R1 (TP2) to CT R1 (TP23) and from CX R1 (TP2) to CX S2 (TP4). Connect a floated oscilloscope with the single lead at CT R2 (TP24) and the return at CT R1 (TP23).

c* Rotate the motor drive around 0 deg. by rotating the slip clutch gear by hand until a null is obtained on the oscilloscope. Connect the second charnel of the oscilloscope to S1 (TP3). Display both waveforms. Rotate the drive in such a way as to produce a counter clockwise rotation of the synchro shaft (as seen by looking at the shaft end). The two waveforms should appear in phase. Using an inspection mirror verify that the mark on the synchro shaft is in alignment with the mark on the synchro housing (this will assure proper synchro zero). Set REF switch to OFF. Disconnect jumpers (except brake power supply) and oscilloscope.

4.7.3.6 Function potentiometer.

a. Position the drive to one of the 10 deg. positions using the synchro of servo drive tester set to 12.71 deg.

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CAUTION: Always approach from null. Set brake power supply to OFF. Connect a jumper between TP13 and TP14 (-15V). Connect dc power supply, item 4, to provide -15 V at TP13 with respect to> 2 RET (TP15). Connect a dc null voltmeter between > 2 SIG (TP16), and λ 2 RET (TP15) and read a voltage of 2.4 ± 0.4 V. Rotate the drive to the other 10 deg. position using the synchro of the servo drive tester set to 347.29 deg. (-12.71 deg.).

CAUTION: Always approach from null. The voltage at this 10 deg. position shall also be 2.4 ± 0.4 V. In addition, the difference between this voltage and the voltage obtained at the other 10 deg. position shall not differ by more than 0.2 v.

b. Remove the alignment pin and rotate the drive to the O deg. position (the null position). A null voltage no greater than 120 mV shall be obtained.

c. Rotate the drive slowly while observing the voltmeter. The **voltage** should **vary smoothly** from null to -15 V and then drop suddenly back to zero at about -25 deg. The same response should be seen when rotating the drive in the opposite direction. Turn off the -15 V supply and disconnect it. Remove the jumper between TP13 and TP14.

4.7.3.7 Microswitch stop.

Connect an ohmmeter between TP1l and TP12. Move traverse servo drive to +27 deg. on traverse alignment fixture. Continue to move traverse servo drive until ohmmeter shows microswitch operation. Stop drive at this position. Aligning fixture pin shall be capable of being inserted into +27 deg. hole to provide locking. Remove aligning fixture pin.

b. Move traverse servo drive to O deg. Continue to move traverse servo drive until ohmmeter shows microswitch operation. Stop drive at this position. Aligning fixture pin shall be **capable** of **being inserted into -27 deg. hole to provide locking.** Remove aligning fixture pin.

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4.7.3.8 Backlash.

a. All antibacklash gears shall be visually inspected using item 8 and shall be loaded (spring, compressed) between two to four teeth. This will provide approximately 10 inch-ounces of preload in the data gear train.

b. When the gear train is deflected to overcome the antibacklash gears and is released the gears shall return to their original position with no evidence of loose gears or mechanical play.

4.7.3.9 Servo drive dynamic test. The traverse servo drive after having met the requirements of 4.7.3 through 4.7.3.8 shall meet the dynamic test as specified in MIL-A-50687 when substituted into an Antenna AS-2049/VPS-2. Antenna AS-2049/VPS-2 and test facilities shall be as specified in MIL-A-50687 (see 3.5.9).

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Packaging Data Sheet PS10548052. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The traverse servo drive provides mechanical movement of the antenna in traverse when used with Radar Set AN/VPS-2. The potentiometer provides lead angle information to the sight current generator, required when solving the fire control problem.

6.2 Acquisition Requirements. Procurement documents shall specify-the 'following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Applicable stock number.

c. First article sample requirements.

d. Packaging requirements, if other than specified in Section 5.

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e. Serialization requirements, if applicable.

f. Certificate of Conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I , Picatinny Arsenal, NJ **07806-5000. This** address will be specified on-the Contract Data Requirements List, DD **Form** 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Radar Set AN/VPS-2
Traverse Servo Drive

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Army - AR

Preparing activity:
Army - AR

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-50666A (AR)	30 APRIL 1990

3. DOCUMENT TITLE

SERVO DRIVE, TRAVERSE: 10548052

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION & SPECIFICATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S/BLDG 6 PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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